

DATA ANALYTICS PORTFOLIO PROJECT

Commercial Card Targeting & Product Analytics

An end-to-end analytics pipeline for prospect targeting, data enrichment, and product experimentation — built for any B2B commercial card or lending business.

Python · SQL · Statistical Testing · Data Visualization



THE CHALLENGE

Growing a card portfolio is hard without reliable data

Commercial card and lending businesses sit on huge amounts of prospect and customer data — but three recurring problems keep that data from driving better decisions.



Messy, incomplete data

Prospect records arrive with missing fields and inconsistent formatting, making it impossible to trust who's actually a good fit.



No prioritization

Every lead gets treated equally, so sales effort and marketing spend go to low-propensity accounts as often as high-value ones.



Guesswork on features

New product features roll out without rigorous evidence that they change customer behavior — or whether they're worth the cost.

THE SOLUTION

A repeatable, five-stage analytics pipeline

Each stage turns raw, imperfect data into a decision the business can act on immediately.



The loop: enrich data → target the right accounts → test product changes → report the impact → repeat.

THE DATA FOUNDATION

A realistic, synthetic dataset — safe to build and share publicly

Every record is synthetically generated — no real customer, prospect, or transaction data is used anywhere in this project — while preserving the statistical patterns of a genuine commercial-card book of business.

6,000

Prospect
records

2,200

Acquired
customers

~40,000

Card
transactions

2,200

A/B test
participants

8 spend categories · 10 industries · 5 regions · 180 days of transaction history · 14 monthly acquisition cohorts

Cleaning messy data, then scoring it for targeting



Data Quality Enrichment

- **Inconsistent text formatting**
Standardized (casing, whitespace)
- **Missing engagement scores**
Imputed via industry + region median
- **Missing revenue estimates**
Imputed via company-size median
- **Duplicate records**
Detected and flagged

~**65%** of prospect records touched by cleaning or imputation



Targeting Readiness Score

A 0–100 composite score blending three firmographic signals:

40%

Digital
Engagement

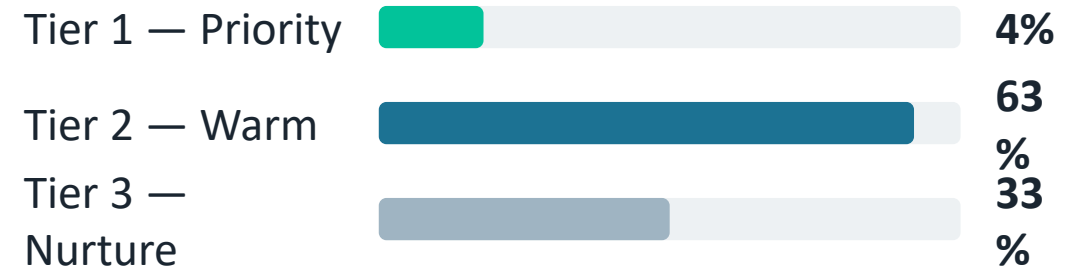
30%

Company Scale

30%

Business
Tenure

Resulting Targeting Tiers



Sales and marketing focus outbound effort on Tier 1 accounts first.

STAGE 3

SQL turns clean data into ready-to-use business queries

Seven production-style queries answer the questions sales, marketing, and portfolio teams ask every week.



Priority prospect list

Highest-readiness accounts with no existing relationship, ranked for outbound



Industry funnel summary

Tier-1 counts and average readiness by industry



Spend by category

Active customers and total spend per category, last 90 days



Monthly spend trend

Portfolio-level spend trajectory over time



Top-decile customers

Highest-spend accounts, ranked via SQL window functions



A/B & cohort summaries

Feature-test results sliced by acquisition cohort

STAGE 4

Testing a new product feature with statistical rigor

A controlled experiment (control vs. treatment) evaluates a new engagement feature before committing to a full rollout.

Week-4 Activation Rate

Control

42.0%



Treatment

50.8%

+8.8pp

Welch's t-test, $p < 0.001$ — statistically significant

30-Day Dispute Rate

Control

6.0%



Treatment

4.8%

-1.2pp

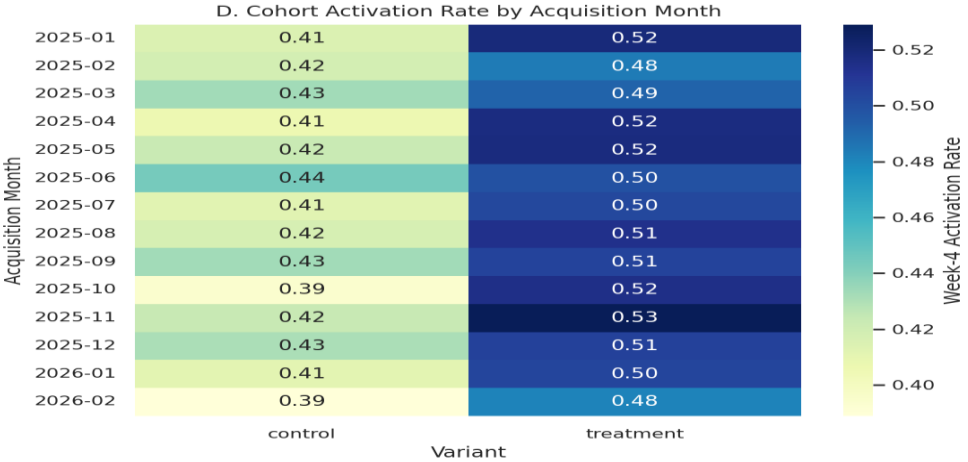
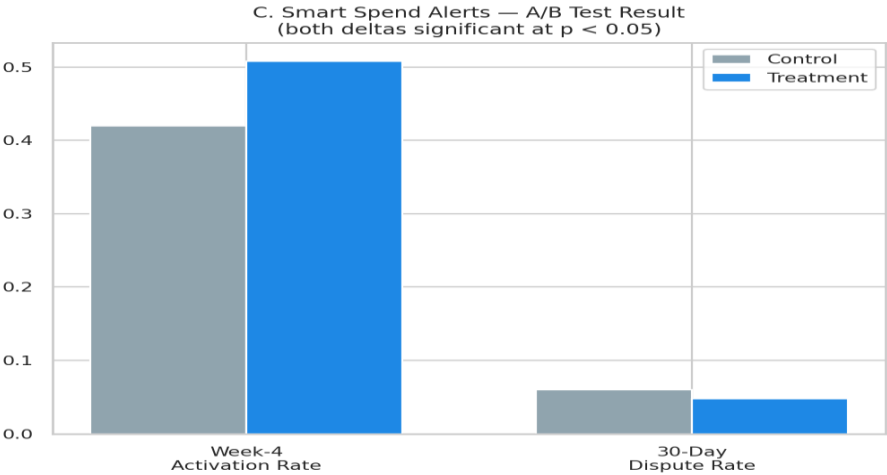
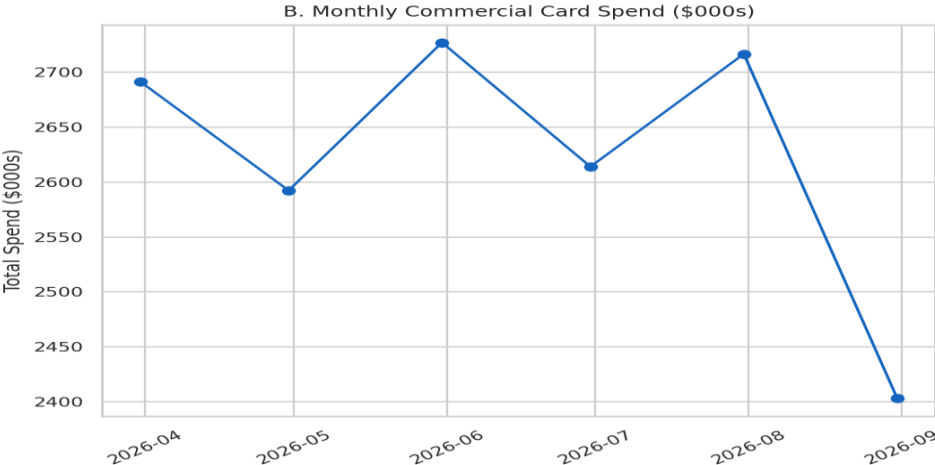
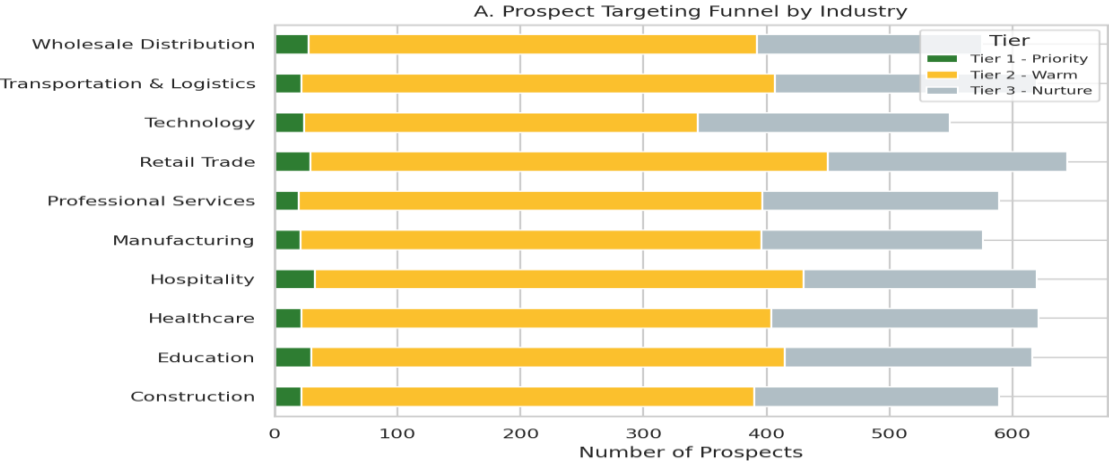
Welch's t-test, $p < 0.001$ — statistically significant



Cohort-validated: the activation lift held consistently across all 14 monthly acquisition cohorts — the effect isn't a seasonal fluke, it's a real, reliable feature impact.

One dashboard, every stakeholder question answered

Commercial Card Product & Targeting Health Dashboard



What the business does differently as a result



Lower acquisition cost

Outbound spend concentrates on the ~4% of prospects most likely to convert, improving CAC and sales-cycle efficiency.



Lower risk & servicing cost

A validated feature reduces the 30-day dispute rate, cutting downstream servicing and fraud-review costs.



Better customer experience

Higher activation means more customers are actively engaging with the product they were sold.



Faster stakeholder reporting

A single dashboard replaces ad-hoc manual reporting, freeing analyst time for the next question.

Recommended next steps: prioritize Tier-1 outbound, fund the full feature rollout, and refresh the enrichment pipeline on every new data cycle.

KEY RESULTS

A repeatable loop for smarter growth

251

priority-tier prospects
surfaced from 6,000

+8.8pp

activation lift from
the tested feature

-1.2pp

dispute-rate reduction
($p < 0.001$)

5

analytics stages,
one seamless workflow

This project demonstrates a repeatable analytics loop — enrich data, target the right accounts, test product changes with rigor, and report the impact — applicable to any commercial card, lending, or B2B subscription business managing a growing customer portfolio.

Built with Python (pandas, numpy, scipy, matplotlib, seaborn) and SQL

Thank you